Math 477 Homework 8

Spring 2026
Rutgers University

May 3, 2026

Question 1: Notice that $X \sim Poisson(2)$ and $Y \sim Binom(10, \frac{3}{4})$.

Computing each term, for X:

$$P(X=0) = \frac{e^{-2}2^0}{0!} = e^{-2}$$
$$P(X=1) = \frac{e^{-2}2^1}{1!} = 2e^{-2}$$
$$P(X=2) = \frac{e^{-2}2^2}{2!} = 2e^{-2}$$

for Y:

$$P(Y=0) = (\frac{1}{4})^{10} = \frac{1}{4^{10}}$$
$$P(Y=1) = \binom{10}{1}(\frac{3}{4})(\frac{1}{4})^9 = \frac{30}{4^{10}}$$
$$P(Y=2) = \binom{10}{2}(\frac{3}{4})^2(\frac{1}{4})^8 = \frac{405}{4^{10}}$$

Since the random variables are independent we can sum up the combinations for $X+Y=2$:

$$\begin{aligned} P(X+Y=2) &= P(X=2)P(Y=0) + P(X=1)P(Y=1) + P(X=0)P(Y=2) \\ &= \frac{467e^{-2}}{4^{10}} \end{aligned}$$

Question 2: Let $X \sim Poisson(20)$. Thus, $E[X] = Var[X] = 20$.

(a) Using the Markov Inequality, we have

$$P(X \geq 26) \leq \frac{E[X]}{26} = \frac{10}{13}$$

Thus our upper bound is $\frac{10}{13}$.

(b) Using the one-sided Chebyshev Inequality, we have

$$P(X \geq 26) \leq \frac{20}{20+6^2} = \frac{5}{14}$$

Thus, another upper bound for p is $\frac{5}{14}$.

(c) The Z-score is computed to be

$$Z = \frac{26 - 20}{\sqrt{20}} \approx 1.34.$$

Then,

$$p \approx 1 - \Phi(1.34) \approx 0.902.$$